

to provide a pulse to the gate of triac BT136. The availability of the gate pulse is indicated by LEDs. The triac switches the mains supply to the load (25W, 220V bulb). The continuous change of data from 8255 ports forms a different pattern of running lights.

When PA0 of PPI 8255 is high, transistor T1 provides the gate pulse (indicated by LED1). The gate pulse triggers Triac 1 and bulb B1 glows. All of the remaining bulbs (B2 through B24) glow in the same way.

Fig. 2 shows the circuit of the power supply unit. 230V, 50Hz AC mains

is stepped down by transformer X1 to deliver a secondary output of 9V, 500 mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC 7805 (IC5). Capacitor C2 bypasses ripples present in the regulated supply.

Construction

An actual-size, single-side PCB for microprocessor-based chaser light circuit is shown in Fig. 3 and its component layout in Fig. 4. Suitable connectors are provided on the PCB for PPI 8255

of the microprocessor kit and the bulb. Assembling the circuit on a PCB minimises time and assembly errors. Carefully assemble the compo-

nents and double-check for any error. Connect the assembled circuit to the PPI 8255 connector available in the microprocessor kit. Use a heat-sink with regulator IC 7805. For triac, a heat-sink is necessary when the power consumption per triac exceeds 60 watts.

Software

The source program is written in Assembly language using instruction set mnemonics of 8085 microprocessor. It is self-explanatory, as suitable comments have been added wherever required. Trial version of 8085 simulator IDE is available from Oshon Software and it can be downloaded from www.oshonsoft.com/downloads.html website.

First, initialise the stack pointer and configure the PPI 8255 as output by control word 80H. When the 8085 program for chaser light is executed in the kit, the display shows "E - CODE" (enter the code). Press 'next'

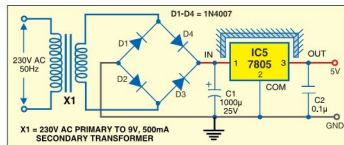


Fig. 2: Power supply

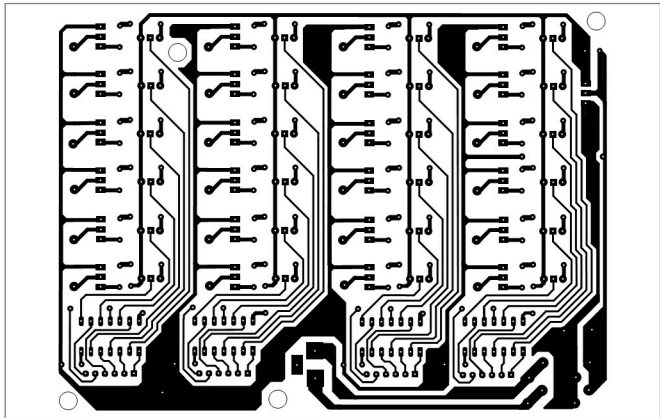


Fig. 3: An actual-size, single-side PCB for microprocessor-based chaser light